

Grade Eight Science - Statics Practice Problem Set 2

MYP Command Terms that are used on this problem set:

State: Give a specific name, value, or other brief answer without explanation of calculator	Outline: Give a brief account or summary
Apply: Use knowledge and understanding in response to a given situation	Describe: Give a detailed account or picture of a situation, event, pattern, or process
Recall: Remember or recognize from prior learning experience	Construct: Display information in a diagrammatic or logical form
Analyze: To identify parts and relationships, and to interpret information to reach conclusions	Interpret: Use knowledge and understanding to recognize trends and draw conclusions from given information

Static Electricity Practice Problems

1. **Recall** that the type of static charging used between the balloon and the hair is called charging by: (Level 1-2)
 A. Conduction
 B. Grounding
 C. Friction
 D. Induction
 E. Both B and D
 F. Both A and C
2. A rubber balloon is rubbed against someone's hair. The balloon is then brought near a rod and the balloon is repelled by the rod.
- A. After the balloon and the hair were rubbed together, **state** the charge left on each item? **Outline** your reasoning below. (Level 3-4)

Since hair has a lower electron affinity than rubber, the hair will put electrons on the rubber balloon. This makes the hair positive since it lost negative electrons and the rubber balloon negative since it gained negative electrons

- B. **State** the charge left on the rod and **outline** your reasoning below (Level 3-4)

Since the balloon is now negative, as outlined in B above, the rod must also be negative because in order for two objects to repel one another they must have the same charge. There is no neutrality in repulsion. Both objects must have the same charge in order to repel.

3. Positive ball X is brought near neutral ball Y. After ball X is brought near ball Y, ball X and ball Y are touched together.
- A. **Outline** what happens to the charges on ball Y when ball X is brought near it? **State** the type of charging used. **Outline** your reasoning below? (Level 5-6)

When ball X is brought near neutral ball Y, the charges on ball Y will polarize. Because ball X is positive, the negative charges on ball Y will move close to the positive ball X, while the positive charges on ball Y will move as far away from ball X as they can since they are repelled.

What's happening here is just polarization. The balls don't change charges since there was no contact between them yet.

- B. Once ball X and ball Y come into contact with one another **state** the charges left on ball X and ball Y? **Outline** your reasoning below. **State** the type of charging used and **state** if it leaves a permanent charge. (Level 3-4 and Level 5-6)

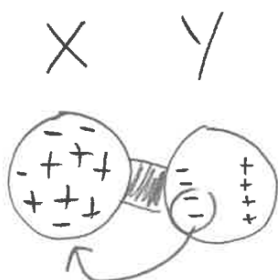
Once balls X and Y are touched together, ball X sees it as a chance to become neutral. ball X takes electrons from ball Y in order to ground itself to get back to neutral. Because ball Y lost electrons ball Y is now positive.

This type of charging is called conduction.

The charge left on ball Y is permanent because there was contact and it lost electrons to ball X; ball Y will remain positive until you ground it.

Ball X is now neutral and it will remain neutral until it is charged using another material at a different time

- C. **Apply** scientific knowledge to **construct** a diagram of the charging described in this question 2B above (Level 5-6)



4. A positive plastic rod is brought near a neutral electroscope.
- A. **Outline** how you can charge a plastic rod positively and **construct** a diagram of the charging that shows the movement of the electrons from one object to another. (Level 5-6)

Outline	Construct
A neutral plastic rod can be charged positively by rubbing it against a material with a higher electron affinity such as polyester (or any material above plastic on the Triboelectric series) Since plastic has a lower electron affinity than polyester, when they are rubbed together applying friction causes electrons to move from the plastic onto the polyester causing the rod to be left positive since it lost electrons and the polyester to be left negative since it gained electrons	

- B. At the same time a positive rod is brought near the neutral electroscope, a hand is placed on the electroscope. **State** the type of static charging described here? (Level 3-4)

Permanent induction (or induction with grounding) is the type of charging that is used here

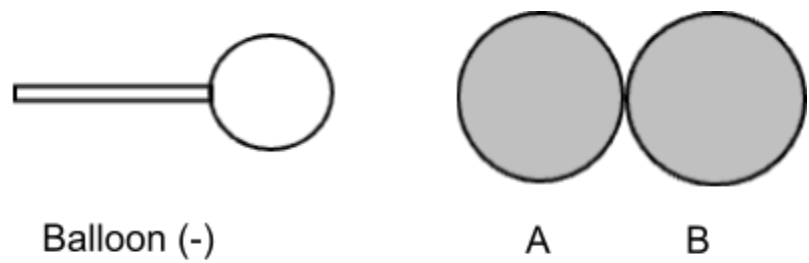
C. Apply scientific knowledge to **construct** the diagrams of the charging indicated in the headings below (Level 5-6)

Neutral Electroscope	Charging of the electroscope with the rod and hand	The electroscope after charging once the rod and hand have been removed

5. A piece of styrofoam is rubbed with fur. The styrofoam is then brought close to the ball of a neutral electroscope and repulsion is observed in the leaf. **State** the charge on the styrofoam, the fur, and the electroscope and **describe** your reasoning for the charge that you assigned to each item (Level 7-8)

Styrofoam	Fur	Electroscope
The styrofoam is negative because it was rubbed with fur. Fur has a lower electron affinity than styrofoam so when they are rubbed together the fur will transfer electrons onto the styrofoam making the styrofoam negative and the fur positive	The styrofoam is negative because it was rubbed with fur. Fur has a lower electron affinity than styrofoam so when they are rubbed together the fur will transfer electrons onto the styrofoam making the styrofoam negative and the fur positive	When the styrofoam is brought near the ball of a neutral electroscope it causes the charges on the electroscope to polarize as an electric field is created around the electroscope. Because the styrofoam is negative, the positive charges on the electroscope are attracted to the styrofoam, so they come up to the ball of the electroscope to be near the styrofoam. The negative charges on the electroscope are repelled by the styrofoam ball, so they move down to the leaf of the electroscope allowing you to see repulsion in the leaf

6. Neutral ball A is in contact with neutral ball B. A negative balloon is brought near ball A. **Interpret** the information given and **apply** scientific knowledge about static charging to answer the following questions about this scenario.



Once ball A and ball B have been separated what charges do they have? What charge is left on the balloon? **Describe** your reasoning below. (Level 7-8)

Balloon	Ball A	Ball B
The balloon remains negative since it has not been in contact with anything. Charges can only transfer with contact between objects	The charges on Ball A will polarize when the balloon is brought near. Since the balloon is negative, the positive charges on Ball A are attracted to the balloon and will therefore move close to the balloon (to the left side of Ball A). The negative charges on Ball A are repelled by the balloon and will therefore move to the right side of Ball A as far away from the balloon as they can get Ball A remains neutral because it was only in contact with Ball B, another neutral object	The polarization of Ball A causes Ball B to polarize. The negative charges on the right side of Ball A attract the positive charges on Ball B causing them to move to the left side of Ball B closest to the negatives on Ball A. The negative charges on Ball B are repelled by the negative charges on Ball A so the negatives on Ball B move to the right side of Ball B, as far away from Ball A as they can get. Ball B remains neutral because it was only in contact with Ball A, another neutral object

7. This question involves a negative ebonite rod and four electroscopes. Electroscope 1 is neutral; the charges on the other three electroscopes are unknown
- A. A negative ebonite rod is brought into contact with electroscope 1. **State** the type of charging and **state** the charges are left on the rod and electroscope 1 after contact. **Outline** your reasoning below (Level 3-4)

The type of charging is conduction because there is contact without the objects being rubbed together. Since there is contact, the rod gives electrons to the electroscope to try and become neutral.

Because the rod loses electrons it is now neutral because it gave away electrons to the electroscope in order to ground itself. The electroscope is now negatively charged because it gained electrons from the rod.

- B. Electroscopes 1, 2, 3, and 4 are brought near one another. Electroscopes 1, 2, and 3, are all attracted to electroscope 4. Electroscopes 2 and 3 have no effect on each other. **Analyze** this information in order to **state** the charge on each electroscope and **describe** your reasoning below. (Level 7-8)

Electroscope 1 is negative because it was charged negatively using conduction with a negative ebonite rod in question A above (see the answer to A above)

Electroscopes 2 and 3 are neutral. The only way they could have no impact on each other is if neither electroscope had a charge.

Electroscope 4 must be charged because two neutral objects are attracted to it. Because Electroscope 1 is negative and it attracts electroscope 4, which we know is not neutral, Electroscope 4 must have a positive charge.

8. **Interpret** the following information to **state** the charge left on each ball and **describe** your reasoning for the charge given (Level 7-8)

- Ping pong balls A, B, and D are all attracted to one another.
- Ball C repels ball B, and
- Ball A has a negative charge.

Ball	Ball A	Ball B	Ball C	Ball D
Charge	Negative	Positive	Positive	Neutral
Reasoning	This charge was given in the question information	Ball B must have a charge that is opposite to Ball C- neutral objects can't repel, so B can't be neutral Since Ball A is negative, and Ball B must be charged, Ball B must be positive because opposites charges attract	Ball C must have a charge that is opposite to Ball B - neutral objects can't repel, so C can't be neutral Since Ball B is positive, Ball C must also be positive because like charges repel	Ball D is neutral. Ball D attracts Ball A, which is negative, and Ball B, which is positive. Since D attracts both a negative object and a positive object it has to be neutral

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